INTERNSHIP REPORT

on

Development of An AI Based Chat Bot for Data Regression Analysis and Exploration of Python Based Aerospace Applications

A Report submitted in partial fulfillment of the

Requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

INFORMATION TECHNOLOGY

Submitted by,

Chillarge Sai Dedeepya (160622737078)

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Syeda Viquar Sultana (160622737120)

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Under the Guidance of

Dr. Jagadish Babu

Scientist, 'E', DRDL, Hyderabad.

Duration: 15th April, 2025 to 15th June, 2025



Department of Information Technology

STANLEY COLLEGE OF ENGINEERING AND TECHNOLOGY FOR WOMEN (AUTONOMOUS)

Chapel Road, Abids, Hyderabad – 500 001

(Affiliated to Osmania University & Approved by AICTE) (All eligible UG Courses are accredited by NBA & Accredited by NAAC with 'A' Grade) [2024-2025]

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CERTIFICATE

This is to certify that the "Internship report" on Development of An AI Based Chat Bot for Data Regression Analysis and Exploration of Python Based Aerospace Applications submitted by **Ms. Chillarge Sai Dedeepya (160622737078), Ms. Sunerah Hasmitha Jaiswal (160622737119), Ms. Syeda Viquar Sultana (160622737120), Ms. Laxmi Manasa Tumuluri (160622737121)** is work done by them and submitted during 2024 – 2025 academic year, in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Information Technology, at DRDL, Hyderabad.

Internship Supervisor

Dr. Jagadish Babu

Scientist 'E'

DRDL, Hyderabad.

Head of Department

Dr. B. Srinivasu

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Government of India, Ministry of Defence
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Defence Research & Development Organisation
उन्नत प्रणाली प्रयोगशाला
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PO Kanchanbagh, Hyderabad - 500 058

No. ASL/23/2025/3340/PW/18/28

Date: 1st April 2025

To

The Principal
Stanley College of Engineering and
Technology for Women (Autonomous)
Chapel Road, Abids,
Hyderabad - 500001

SUB: - PERMISSION FOR INTERNSHIP

Ref: Letter dated 28th Mar' 2025.

1. Following B.Tech (IT) 3rd year students of your college have been permitted to undergo Major project work in our Laboratory for a period of **from 15th April to 15th June 2025. Subject to submission of satisfactory Police Verification Certificate.**

Sl. NO.	Name	Roll NO.
1	Ms. Chillarige Sai Dedeepya	160622737078
2	Ms. Sunerah Hasmitha Jaiswal	160622737119
3	Ms. Syeda Viquar Sultana	160622737120
4	Ms. Laxmi Manasa Tumuluri	160622737121

2. They should be in possession of satisfactory police verification letter and three – passport size color photographs for issue of temporary pass prior to commencement of training. Documents pertaining to ID-Proof and Address-Proof are also required. You may advise the students to meet Head HRDG, ASL for necessary advice.

(Hemant Kumar)
Scientist 'E'
Head, HRDG, ASL

हेमंत कुमार / HEMANT KUMAR
प्रधान, एच.आर.डी.जी. / Head, HRDG
ए.एस.एल., हैदराबाद-58. / ASL, Hyderabad-58.

Encl: Suggested Format for Police Verification (Overleaf)

OVERVIEW OF INTERNSHIP ACTIVITY

WEEK – 1

Date	Details of the Internship Activity Performed/ Completed
15/04/2025 to 22/04/2025	✓ Overview of Rocketry and Missile Types, Rocket Propulsion Principles and Design ✓ Missile Launching, Guidance, and Control ✓ Introduction to Python and Programming Basics ✓ Core Python Data Types and Control Structures ✓ Python Functions and Operations ✓ Studied data science fundamentals from Python Data Science Handbook. ✓ Explored core Python libraries: NumPy, Pandas, Matplotlib, and Scikitlearn. ✓ Worked with Jupyter and IPython for interactive coding and analysis. ✓ Studied core Python concepts, including data types, functions, and object oriented programming. ✓ Practiced coding challenges and algorithmic problems like sorting,searching, and data structure implementation. ✓ Explored key Python libraries for data science (NumPy, Pandas, Matplotlib) and machine learning (Scikit-learn, TensorFlow).

WEEK – 2

Date	Details of the Internship Activity Performed/ Completed
23/04/2025 to 29/04/2025	✓ Tactical Missiles, Strategic Missiles, Rocket Equations, Cures, Scope, Sky, Types of Missiles, Guidance Keys, Ballistic Missiles, Phases of Flight, Ballistic Trajectory, Tactical Missiles and its Guidance Schemes, Types of Tactical Missiles, Strategic Missiles and their Guidance Schemes, Types of Strategic Missiles, Importance of Escape Velocity in Rockets and Missiles. ✓ Regression models – Linear, Logarithmic, L1, L2 norms, Least square methods, Gaussian distribution, Normal distribution. ✓ Software Tools: AutoCAD,Python in AutoCAD, ANSYS 2022. ✓ Simple implementations of logarithms, minima, integration & differentiation, numerical methods, Plotting of functions using Python, Gaussian & other distributions (Maths), Applications of determinants. ✓ Laplacian transform, Z-transform, Signal processing, Linearity, Continuous & discrete variables, random variables, Fourier transform, inverse Fourier transform.

WEEK – 3

Date	Details of the Internship Activity Performed/ Completed
30/04/2025 to 06/05/2025	✓ Missile Defense and Interception, Single Stage Missiles, Multi-Stage Missiles, Rocket Motor and Machine Learning, Missile Sizing Overview and Mission Requirements, Estimating Thrust and Burn Time, Mass Breakdown, Why We Require Multi-Stage Rockets, Types of Rockets, Purposes of Different Rockets and their Guidance Keys along with Real- Life Examples, Aircraft Propulsion Types. ✓ Plotting all types of distributions using Python, accuracy, precision & recall metrics. ✓ Software Tools: MATLAB, AI Tools like DeepSeek. ✓ Applications of dot & cross product, Vector calculus & its theorems, Mean, variance, standard deviation, skewness, scalars, vectors & tensors, Newton's laws, Trajectory of freely falling object. ✓ Importance of trigonometric functions, Transfer functions, Cross- correlation & auto-correlation, Orthogonality, Sigmoid function.

WEEK – 4

Date	Details of the Internship Activity Performed/ Completed
07/05/2025 to 13/05/2025	✓ Bottle Rocket Drawing via ezdxf ✓ Bug Fixes in Text and Dimensions ✓ AutoCAD Automation with COM API ✓ Automated Detection of Data Distribution Patterns ✓ Peak and Amplitude Identification in Data Series ✓ Regression Pattern Analysis ✓ Plotting final velocity changes v/s the number of stages in a rocket ✓ AutoCAD Rocket Drawing (Mistakes and fixes) ✓ Monte-Carlo Method (1 variable and 2 variables), Skewness, kurtosis ✓ Consider a function and study the effect of changes of variables on the function. ✓ What happens when you apply L1 and L2 norms? ✓ Write a python code to find roots of quadratic equation $AX^2+BX+C=0$ and polynomial $AX^4+BX^2+C=0$. ✓ Given 1D, 2D or 3D data, analyse the data and identify the distribution it comes under and plot the distribution too and then define its behaviour, also identify the peaks and amplitude in the data and give the equation/general equation for it. Identify the regression pattern it might follow or the type of regression we can employ on it to understand more about the given data. ✓ FreeCAD application of fully dimensioned 3D missile model ✓ 2D Drawing with TechDraw, design of Missile using FreeCAD.

WEEK – 5

Date	Details of the Internship Activity Performed/ Completed
14/05/2025 to 20/05/2025	✓ Trained and evaluated multiple machine learning models: Linear Regression, Ridge Regression, Lasso Regression, SVR, AdaBoost, KNN, and XGBoost. ✓ Used consistent performance metrics such as R^2, MAE, and RMSE to evaluate models. ✓ Visualized regression surfaces and plotted actual vs predicted values using matplotlib, seaborn, and Plotly. ✓ Compared performance results and finalized the best model based on accuracy, interpretability, and speed. ✓ Designed and developed a chatbot-style frontend UI using Vercel AI tools. ✓ Implemented file upload functionality and animated analysis option buttons/dropdowns. ✓ Integrated themed layout with smooth CSS transitions and user-friendly interactions.

WEEK – 6

Date	Details of the Internship Activity Performed/ Completed
21/05/2025 to 27/05/2025	✓ Developed backend using FastAPI with key endpoints: /upload-data, /analyze, /generate-plot. ✓ Embedded hardcoded, rule-based chatbot responses to guide users through each analysis step. ✓ Implemented logic-based responses like “Please upload a file” and “Select an analysis type”. ✓ Integrated frontend and backend via REST APIs and tested complete data flow. ✓ Verified responses for all analysis types and visual output consistency. ✓ Added error handling, loading indicators, and responsive design elements. ✓ Conducted end-to-end testing with CSV files to validate system robustness.

WEEK – 7

Date	Details of the Internship Activity Performed/ Completed
28/05/2025 to 06/06/2025	✓ Compiled technical content from Weeks 1–6 into a structured report format. ✓ Documented the objective, methodology, and outcomes of all machine learning models used. ✓ Included detailed explanation of regression techniques along with plots and results. ✓ Added dataset descriptions and visualization snapshots. ✓ Highlighted frontend-backend integration steps ✓ Ensured proper citation of tools, libraries, and external references. ✓ Finalized layout with professional formatting, headings, and annexures for supporting materials.

Organization Seal

Signature of the Supervisor

ACKNOWLEDGEMENT

Firstly, we, would like to thank **Dr. Jagadish Babu**, Scientist 'E', DRDL, Hyderabad for giving us the opportunity to do an internship within the organization.

We would like to express our gratitude for the training experience at DRDL, Hyderabad. The learning opportunities provided have been invaluable, helping us enhance our skills and gain practical knowledge. This training allowed us to build confidence in our abilities and contributed significantly to our professional growth.

We are highly indebted to **Principal (Dr. B. L. Raju),** for the facilities provided to accomplish this internship. We would like to thank our Head of the Department, **Dr. B. Srinivasu** for his constructive criticism throughout our internship.

We are extremely great full to our department staff members and friends who helped us in successful completion of this internship.

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ABSTRACT

As a part of our internship at Advanced Systems Laboratory (ASL), we learned the application and then implementation of Python in aerospace-related tasks. We started by understanding the fundamentals of rockets and missiles, including their types, structures, and the role of propulsion, aerodynamics, and guidance systems. Using Python, we implemented programs to explore concepts like multistaging and optimal mass distribution. These helped us understand how multiple stages in a rocket can increase velocity, reduce fuel consumption, and improve the delivery of payloads or warheads.

We also worked on 2D design using AutoCAD. We explored various applications of the software and created basic 2D models of rockets, focusing on both the internal arrangement and external shape. These designs helped us visualize and relate theory to real-world structures, especially during early-stage concept development and structural planning.

A major highlight of our internship was developing an AI-based chatbot application. The chatbot allows users to upload datasets from public sources and perform important data analysis tasks such as Lasso Regression, peak detection, amplitude calculation, and distribution analysis. It explains the theory behind each technique and presents the results in a user-friendly way, making it useful even for those with minimal technical background.

Through this internship, we gained hands-on experience in applying Python programming, AI tools, and design software to real-world aerospace problems. It helped us improve both our technical and analytical skills and showed us how modern technologies can contribute to smarter, faster, and more efficient solutions in the defense and aerospace industries.

ABOUT THE ORGANIZATION

Overview of DRDL (Defence Research and Development Laboratory)

History of DRDL:

The Defence Research and Development Laboratory (DRDL) is a premier institution under the Defence Research and Development Organisation (DRDO), Ministry of Defence, Government of India. Established in 1961 and based in Hyderabad, DRDL was founded with the vision of developing indigenous missile systems for India's defense needs. It has played a crucial role in the nation's self-reliance efforts in missile and aerospace technologies. Over the years, DRDL has been responsible for several successful missile projects, including the development of tactical and strategic missile systems such as the Prithvi, Akash, and Agni series. DRDL remains at the forefront of missile R&D, contributing significantly to India's defense preparedness.

Vision and Mission:

DRDL's vision is to build robust and self-reliant missile systems that ensure national security and strategic superiority. Its mission is to design, develop, and deliver cutting-edge missile technologies and systems, keeping in mind future warfare scenarios and global defense standards. DRDL aims to constantly innovate in propulsion systems, control technologies, warhead design, and system integration, thereby strengthening India's strategic capabilities.

Sector and Services:

Operating within the defense research sector, DRDL specializes in missile technology, including:

1. **Strategic and Tactical Missiles:** Development of short-range, medium-range, and long-range missile systems for land, air, and sea operations.
2. **Guidance and Control Systems:** Advanced control mechanisms for missile navigation, targeting, and precision.
3. **Propulsion Systems:** Innovations in solid, liquid, hybrid, and air-breathing propulsion systems.
4. **Simulation and Testing Facilities:** Use of state-of-the-art laboratories for virtual simulations, structural analysis, environmental testing, and real-time weapon system validation.

5. **Collaborations and Joint Projects:** DRDL collaborates with other DRDO labs, private defense firms, and international partners to drive innovation and technology exchange.

Internship and Learning Opportunities:

DRDL offers internship and research opportunities to students and young professionals in the fields of aerospace, mechanical, electronics, and computer science engineering. Interns are exposed to real-time defense projects and gain hands-on experience in Python programming, simulation tools, 2D/3D modeling, control system design, and AI integration in defense systems. These internships are aimed at nurturing young talent and creating a skilled workforce for India's defense R&D ecosystem.

Customer and Impact:

DRDL's primary customers include the Indian Armed Forces—Army, Navy, and Air Force—as well as strategic partners like Bharat Dynamics Limited (BDL) and Hindustan Aeronautics Limited (HAL). Its work directly supports India's defense preparedness, modernization efforts, and technological advancement in the missile domain. DRDL also plays a key role in supporting Make-in-India and Atmanirbhar Bharat initiatives by reducing dependency on foreign defense imports.

Conclusion:

DRDL stands as a vital pillar in India's defense research framework, driving innovation in missile technology and enhancing national security. Through its advanced R&D, skilled workforce, and focus on self-reliance, DRDL continues to push the boundaries of indigenous defense capabilities. Its internship programs and student engagements also contribute to the development of the next generation of defense scientists and engineers.

TECHNICAL OVERVIEW

Rockets And Missiles

Rocket: A rocket is a vehicle that moves by pushing gas out from its back. It is powered by fuel but usually **doesn't have control or tracking** once it's launched. It just goes where it's aimed.

Example: A firework or the rocket that sends satellites into space.

Missile: A missile is a **guided rocket** — it can **find and follow a target**. It has sensors and sometimes even a small "computer" to adjust its path.

Types of Rockets and Missiles

By Guidance (How It Moves and Finds Target)

1. Unguided Rocket ("Dumb" rocket)

- No brain. Goes in a straight line.
- Cheap but less accurate.
- Example: **Artillery rockets**, fireworks.

2. Guided Missile ("Smart" missile)

- Has tracking systems like radar, GPS, or heat sensors.
- Can change direction mid-air to hit the target.

By Launch Place (Where It Starts From)

1. Surface-to-Surface Missiles

- Launched from land to hit land.
- **Examples**: Agni, Prithvi (India), Scud (Iraq).

2. Surface-to-Air Missiles (SAM)

- Launched from the ground to hit **aircraft or enemy missiles**.
- **Examples**: Akash (India), S-400 (Russia), Patriot (USA).

3. Air-to-Air Missiles (AAM)

- Fired from one plane to destroy another.
- Fast and accurate.
- **Examples**: AIM-9 Sidewinder, Astra (India).

4. Air-to-Surface Missiles (ASM)

- Dropped from aircraft to destroy tanks, buildings, etc.
- **Examples**: Hellfire (USA), BrahMos-A (India).

5. Ship/Submarine-Launched Missiles

- Fired from navy ships or submarines.
- Can hit sea or land targets.
- **Examples**: Tomahawk, K-15.

Classification Of Missiles

1. Ballistic Missiles

- **Function**: Fly into space (like a big arc) and fall back down on the target.
- **Use**: Long-range attacks, often carrying powerful warheads.
- **Range Types**:
    - **Short-range (SRBM)**: up to 1,000 km (e.g., Prithvi)
    - **Medium-range (MRBM)**: 1,000–3,500 km (e.g., Agni-II)
    - **Intermediate-range (IRBM)**: 3,500–5,500 km
    - **Intercontinental (ICBM)**: over 5,500 km (e.g., Agni-V, Minuteman III)
- **Example**: Agni series (India), Trident (USA)

2. Cruise Missiles

- **Function**: Fly low like an airplane, staying close to the ground to avoid radar.

- **Use**: Precision attacks on buildings, bunkers, or ships.
- **Speed Types**:
    - **Subsonic**: Slower than sound (e.g., Tomahawk)
    - **Supersonic**: Faster than sound (e.g., BrahMos)
- **Example**: BrahMos, Nirbhay, Tomahawk

3. Anti-Tank Guided Missiles (ATGM)

- **Function**: Destroy armored vehicles or tanks.
- **Use**: Carried by soldiers or mounted on vehicles/helicopters.
- **Example**: Javelin (USA), Spike (Israel), Nag (India)

4. Anti-Ship Missiles

- **Function**: Target and destroy enemy warships.
- **Use**: Launched from land, aircraft, ships, or submarines.
- **Example**: Harpoon (USA), Exocet (France), BrahMos (India)

5. Anti-Radiation Missiles (ARM)

- **Function**: Find and destroy enemy radars or communication systems.
- **Use**: Suppress enemy air defenses.
- **Example**: AGM-88 HARM (USA), Rudram-1 (India)

6. Anti-Satellite Missiles (ASAT)

- **Function**: Hit and destroy satellites in space.
- **Use**: Disable enemy surveillance or GPS.
- **Example**: Mission Shakti (India), SC-19 (China)

7. Hypersonic Missiles

- **Function**: Travel at very high speeds (over Mach 5 = 6,000+ km/h).
- **Use**: Fast strike, hard to detect or stop.

- **Types**:
    - **Hypersonic Glide Vehicles (HGV)**: Boosted to high altitude and then glide.
    - **Hypersonic Cruise Missiles**: Use air-breathing engines.
- **Example**: Russia's Kinzhal, China's DF-ZF

How Do Missiles Know Where to Go? (Guidance Types)

- **Heat-seeking**: Follows the heat of enemy engines.
- **Radar-guided**: Uses radio waves to track targets.
- **GPS-guided**: Uses satellites to guide itself.
- **Laser-guided**: Follows a laser beam pointed at the target.
- **Wire-guided**: Controlled by wires during flight (used in short-range anti-tank missiles).

Components Of Rockets And Missiles

1. Propulsion System

Purpose: This is the engine that makes the rocket or missile move forward by pushing gases out from the back (thrust).
It decides how fast and how far it will go.

Parts:

- **Engine**: Converts fuel into thrust.
- **Fuel/Propellant**:
    - **Solid fuel**: Pre-packed, ignites quickly, used in most missiles.
        - Example: Agni missile series
    - **Liquid fuel**: Stored separately, mixed during launch, used in rockets and some long-range missiles.
        - Example: ISRO's GSLV rocket
- **Booster**: Extra engine for takeoff in some rockets or missiles. Detaches after use.

2. Warhead *(only in missiles)*

Purpose: This is the part that carries the explosive or destructive load. It damages or destroys the target.

Types:

- **Explosive (High Explosive - HE)**: Causes a blast.
- **Fragmentation**: Spreads small pieces like shrapnel.
- **Penetrating**: Designed to pierce armor or bunkers.
- **Nuclear**: Used for maximum destruction.

3. Guidance System

Purpose: Helps the missile stay on track and find its target. It is what makes the missile "smart".

Types of guidance:

- **Inertial (INS)**: Uses internal sensors (good for short-to-medium range).
- **GPS**: Uses satellite signals to stay accurate.
- **Radar-seeking**: Tracks enemy radar.
- **Infrared/Heat-seeking**: Follows the heat from planes or engines.
- **Laser-guided**: Follows a laser beam pointed at the target.

4. Control System

Purpose: Helps change direction during flight. It follows instructions from the guidance system and moves parts to steer the missile.

Main parts:

- **Fins/Wings**: Adjust flight path like airplane flaps, **Actuators**: Tiny motors that move fins.

5. Airframe (Structure)

Purpose: It's the body or skeleton of the rocket or missile. It holds everything inside and protects it during flight.

Includes:

- **Nose cone**: Front part that helps cut through air.
- **Midsection**: Holds the fuel tanks, engine, electronics.
- **Tail**: May hold fins or nozzles for control.

6. Payload *(in rockets only)*

Purpose: Instead of a warhead, rockets carry useful items like:

- **Satellites**
- **Scientific instruments**
- **Space probes or crew modules**

Summary:

Component	**Function**	**Found In**	**Example**
Propulsion	Provides thrust to move forward	Rockets & Missiles	Agni engine, Falcon 9 rocket
Warhead	Causes damage or destruction	Missiles only	BrahMos warhead, Nag warhead
Guidance System	Finds and follows the target	Guided Missiles only	Astra IR guidance, GPS system
Control System	Steers missile during flight	Guided Missiles only	Akash missile fin actuators
Airframe	Holds and protects all components	Rockets & Missiles	Missile body, nose cone
Payload	Carries useful cargo (e.g., satellites)	Space Rockets only	ISRO's satellite on GSLV

Understanding Multi-Staging in Rockets

Overview:

Multistage rockets are essential for sending satellites and spacecraft into orbit. The idea behind staging is to improve efficiency by shedding unnecessary weight during the flight, which allows the rocket to travel faster and higher. This explains the concept of multi-staging, its advantages, how to determine the best number of stages, and how stage mass should be distributed.

What is a Multistage Rocket?

A multistage rocket is composed of several sections or "stages", each with its own engine and fuel. After a stage uses up its fuel, it is detached, reducing the weight and allowing the remaining rocket to accelerate more efficiently.

Benefits of Multi-staging:

- Reduces mass during flight.
- Increases final speed
- More efficient fuel usage
- Helps reach orbital velocity.

Why Multi-staging Is Needed:

A rocket needs to reach extremely high speeds to escape Earth's gravity. However, a rocket is heaviest at launch because it carries fuel, structure, and payload all at once. Carrying all of that weight to orbit is inefficient.

Problems with Single-Stage Rockets:

- Must carry unused fuel and heavy structure throughout the flight.
- Often can't reach the required orbital speed for satellite launches.
- Wastes energy by accelerating parts that are no longer useful.

How Multi-staging Works:

A multistage rocket separates its structure into multiple parts, called stages, each with its own engine and fuel. After one stage uses its fuel, it detaches and falls away, lightening the load for the next stage. Each subsequent stage then ignites and propels the rocket further, now carrying

less weight. This step-by-step process allows the rocket to gain more speed efficiently. By discarding empty stages, the rocket avoids wasting energy on carrying unnecessary mass. In a multistage rocket:

- Each stage contains its own engine and fuel.
- After the fuel in a stage is used up, the stage is detached
- The next stage ignites and continues, now with less weight to carry. This process increases the overall speed and efficiency of the rocket.

Advantages of Multi-staging: Multi-staging gives many benefits to rocket flights.

- More speed: Each stage helps the rocket go faster.
- Saves fuel: Lighter weight means better use of fuel.
- Flexible use: Stages can be made for different jobs like launching, reaching space, or changing direction.

Choosing the Right Number of Stages: More stages can help a rocket go faster, but having too many stages can make the rocket more expensive and harder to build. So, choosing the right number of stages is important.

- 2 stages are usually enough to send satellites into space.
- 3 stages are better for longer missions, like going to the Moon.
- 4 or more stages are rarely used unless it's a very special mission.

Optimizing Stage Sizes: Each stage of a rocket must be planned carefully. If one stage is too heavy or too light, the rocket may not work well.

- The first stage should be the biggest because it lifts the rocket off the ground.
- The next stages should be smaller and lighter.
- The weight and fuel of each stage should be planned so nothing is wasted.

This helps the rocket go as far and as fast as possible with the fuel it has. Proper stage sizing also reduces the overall cost of the mission by improving fuel efficiency. Engineers carefully calculate and simulate different designs to find the best balance between weight and fuel for each stage.

Real-Life Relevance:

Many rockets we see in real space missions use this idea.

- India’s PSLV rocket has 4 stages and is used to launch satellites.
- SpaceX’s Falcon 9 has 2 stages, and its first stage can even land back and be reused.
- NASA’s Saturn V, used in Moon missions, had 3 stages.

These examples show that multistaging is a real and important part of rocket design today.

Engineering Insight: The Art Behind Staging

Designing a multistage rocket is not just a mathematical optimization task—it's an art of engineering precision. Each stage is custom-built not only for its own job but also to support what comes next. Engineers must factor in aerodynamics, structural stability, fuel type, ignition timing, and even how the stages separate mid-flight without damaging the payload. Furthermore, every gram saved in the rocket structure can be used to carry more fuel or a heavier satellite. This delicate balancing act between mass, thrust, and timing makes staging one of the most exciting challenges in aerospace design. The final configuration of the rocket reflects not just physics, but a blend of strategic trade-offs that lead to mission success.

Infographic/Table:

Feature	**Single-Stage Rocket**	**Multistage Rocket**
Final Speed	**Limited**	**High**
Efficiency	**Low**	**High**
Cost	**Lower(but less effective)**	**Higher(more efficient)**
Reusability	**Possible, but limited**	**Possible (e.g., Falcon 9 first stage)**
Complexity	**Simple**	**Complex**

What Happens to the Stages After Separation?

When a stage separates, it doesn't stay in space forever.

• Lower stages often fall back to Earth or the ocean.

• Some are burned up in the atmosphere during descent.

• Newer rockets (like SpaceX) land back safely for reuse. This makes staging more environmentally friendly and cost-effective.

Looking Ahead: The Future of Rocket Staging

As space exploration advances, engineers are now exploring reusable multistage systems, hybrid propulsion technologies, and even in-space refueling. Future rockets may include AI-controlled staging systems that adjust mid-flight for maximum efficiency. The principle of staging remains central, but how we implement it continues to evolve, offering exciting possibilities for missions to Mars, deep space, and beyond.

Challenges in Multistage Rocket Design:

Even though multi-staging is helpful, it's not easy to design.

Challenges include:

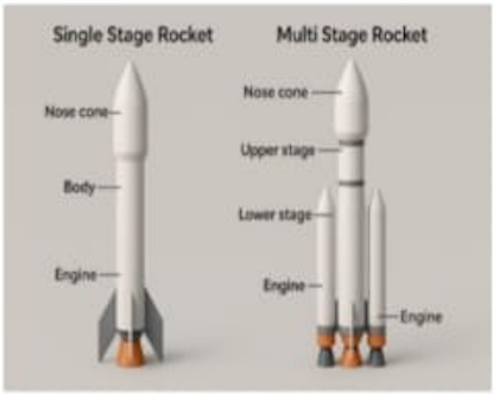
- Timing: Each stage must ignite at the perfect moment.
- Separation safety: Stages must separate smoothly without damaging other parts.
- Weight balance: If one stage is too heavy or light, it affects the whole mission.
- Communication systems: Must switch smoothly from one stage to the next.

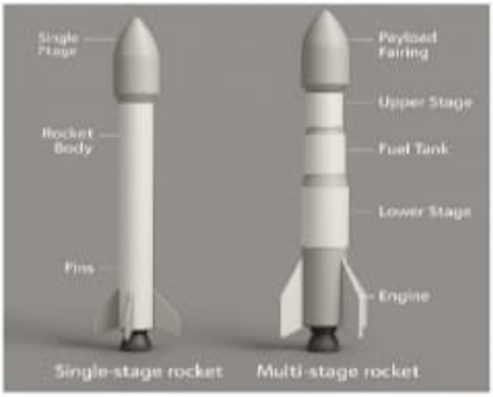
This is why space engineers run thousands of tests before an actual launch.

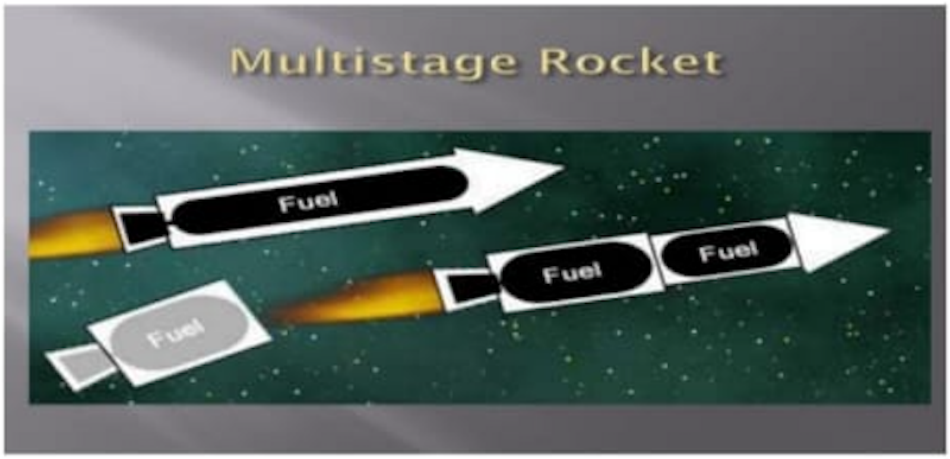
Conclusion:

Multistage rockets are a smart way to reach space. By dropping parts they no longer need, rockets get lighter and move faster. This helps save fuel and go higher. The number of stages and how they are made must be planned carefully. Multi staging is a key idea that makes space travel possible — whether it's sending satellites, going to the Moon, or exploring faraway planets.

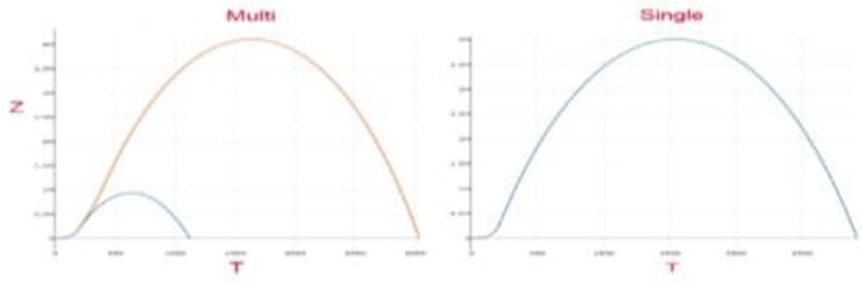
As rocket technology improves, multi staging will continue to play a big role in making space travel faster, safer, and more affordable.



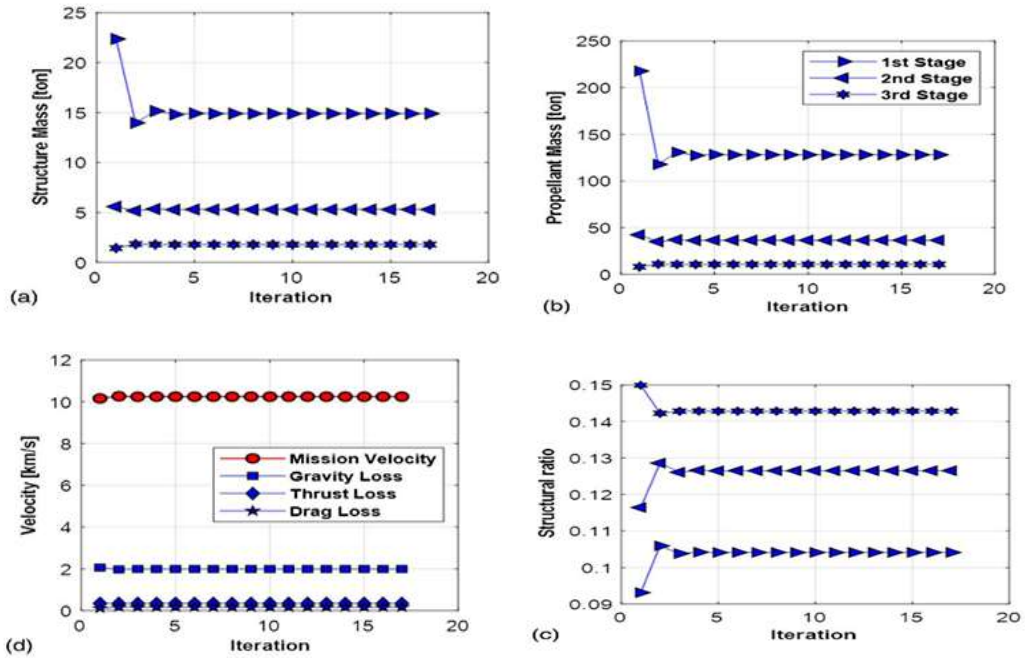




GRAPHS: Stimulation -Developing intuition about altitude and velocity in multi stage rockets.



Multistage Liquid Rocket Weight Estimation and Optimization for Early Design Stages



Python Visualization Report: Multiple Function Plotting

Overview

This Python program demonstrates multiple function plotting using Matplotlib and NumPy libraries to create a comprehensive visualization of three mathematical functions.

Libraries Used

Matplotlib: A plotting library providing object-oriented API for creating visualizations with extensive customization options.

NumPy: Numerical computing library for efficient array operations and mathematical function calculations.

Program Implementation

Data Setup

- Creates x-axis domain from -5 to 9 with 200 data points using np.linspace()
- Establishes the figure size of 10×6 inches for optimal display.

Functions Plotted

1. **sin(x)** - Solid line showing oscillatory trigonometric behavior
2. **cos(x)** - Dashed line displaying phase-shifted oscillation
3. **exp(-x²)** - Dotted line representing Gaussian bell curve with exponential decay

Visualization Features

- Professional styling with distinct line patterns for each function
- Grid overlay for easy value reading
- Legend for function identification
- Proper axis labeling (x, y) and descriptive title
- Output display using plt.show()

Key Benefits

- Clear comparison of different mathematical function behaviors
- Professional presentation suitable for educational and research purposes
- Efficient use of NumPy for high-performance mathematical computations
- Matplotlib's flexibility for customization and styling

Applications

Ideal for mathematical education, engineering analysis, scientific research, and technical documentation requiring multi-function comparisons.

Python Program (Multiple Function Plot):

```
import numpy as np

import matplotlib.pyplot as plt

x = np.linspace(-5, 9, 200)

plt.figure(figsize=(10, 6))

plt.plot(x, np.sin(x), label='sin(x)')

plt.plot(x, np.cos(x), '--', label='cos(x)')
```

```
plt.plot(x, np.exp(-x**2), ':', label='Gaussian')

plt.title('Multiple Function Plot')

plt.xlabel('x')

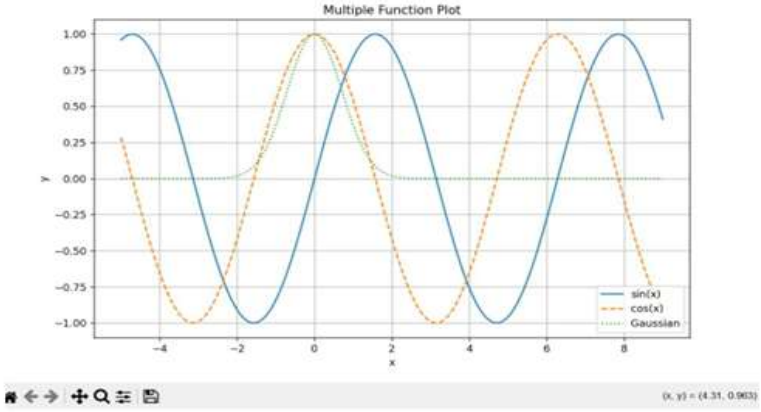
plt.ylabel('y')

plt.legend()

plt.grid(True)

plt.show()
```

Output:



AutoCAD

Executive Summary

This report details AutoCAD automation development through Python programming. Key achievements included bottle rocket design automation, dimensioning solutions, and COM API integration for direct AutoCAD manipulation.

DXF File Generation and Implementation

A comprehensive bottle rocket schematic was successfully created using Python's ezdxf library. The implementation included geometric construction of rocket components (nose cone, body, fins), proper text annotation, and precise dimensioning. Critical measurements incorporated rocket height (76.0 cm), fin width (10 cm), and throat-to-fin distance (7.5 cm). Initial development encountered several technical challenges, particularly with text positioning and alignment methods. The primary issues involved incorrect attribute calls on ezdxf objects,

specifically related to .set_pos() and .set_align() methods. Resolution required implementation of proper positioning using .dxf.insert and alignment through .set_placement() with TextEntityAlignment enumeration.

COM API Integration and Automation

Advanced automation was achieved through win32com.client implementation for direct AutoCAD manipulation. The system incorporated sophisticated layer management across five distinct categories: RocketBody, RocketCurves, Shading, Text, and Dimensions. Complex geometric construction utilized polylines and arcs for rounded features, offset methods for internal contour generation, and ANSI pattern hatching for visual enhancement. Multiple script iterations addressed various technical obstacles including RPC failures, coordinate formatting errors, and method misuse. The final implementation successfully automated complete rocket drawing generation with proper annotation and dimensioning.

Technical Challenges and Solutions

Text Annotation and Alignment Issues

Several critical errors were encountered during initial development phases. AttributeError exceptions occurred with 'Text' objects lacking 'set_pos' and 'set_align' attributes. These stemmed from incorrect method calls on ezdxf objects.Resolution involved replacing deprecated methods with proper positioning techniques using .dxf.insert and implementing alignment through .set_placement() with TextEntityAlignment enumeration.

Dimensioning Implementation

Initial dimensioning attempts encountered DXFTableEntryError due to missing 'EZ_Metric' styles and incorrect method usage such as add_dimension_linear and add_dimlinear. The final approach utilized add_linear_dim() with built-in Standard styles, incorporated proper render() calls for dimension visibility, and implemented optional explode() functionality to convert dimensions into primitive DXF entities.

Drawing Components and Specifications

The final DXF drawing incorporated complete rocket geometry with labeled text for key components. Three primary dimensions were established: rocket height at 76.0 cm, fin width at 10 cm, and throat-to-fin distance at 7.5 cm. AutoCAD-compatible arrows and measurement labels were successfully integrated using standard dimensioning protocols.

Script-Based Automation

Multiple AutoCAD script approaches were developed to address various automation requirements. Script file generation enabled automated drawing creation through .scr file execution within AutoCAD environments. The implementation addressed version compatibility issues and eliminated interactive prompts that previously interrupted automated processes.

Layer Management and Organization

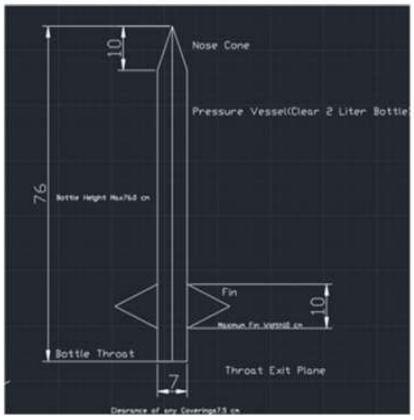
Systematic layer organization was established across five primary categories to maintain drawing clarity and enable selective editing capabilities. RocketBody layers contained primary geometric elements, RocketCurves managed curved features, Shading layers handled visual enhancement elements, Text layers organized annotation components, and Dimensions layers maintained measurement information.

Performance Optimization

Error handling mechanisms were implemented to address common COM interface issues including RPC failures and coordinate formatting problems. Try-except blocks provided safety measures for COM error management, while methodical input validation ensured consistent operation across different AutoCAD versions and configurations.

Technical Outcomes

Development resulted in functional AutoCAD automation tools capable of generating complex technical drawings with minimal manual intervention. The system provides comprehensive dimensioning capabilities, proper text annotation, and sophisticated geometric construction methods with robust error handling across different AutoCAD versions.



FreeCAD

1. Core Features of FreeCAD for Projectile and Vehicle Design

A. Parametric Design

- **Definition**: Parametric modeling lets you define key dimensions (e.g., length, diameter, sweep angle) as variables. You can change the model just by modifying these parameters.
- **Benefit**: You don't need to redraw or remodel from scratch—adjusting one parameter automatically updates the entire model.
- **Application**: In the projectile, the nose cone length, body diameter, and fin size are all adjustable using this method.

B. Modular Architecture

- FreeCAD separates features into **"workbenches"**, each designed for specific tasks. Key workbenches used in this project include:
    - **Part Design Workbench**: For sketch-based solid modeling
    - **Draft Workbench**: For 2D construction and annotations
    - **TechDraw Workbench**: For creating 2D technical drawings with dimensions
    - **Part Workbench**: For combining solids using Boolean operations
- This modularity makes FreeCAD highly **customizable and scalable** for both simple and complex projects.

2. Technical Working

A. Sketching and Modeling

- **Sketching**: You start by creating 2D sketches (e.g., circle for projectile base, triangle for fin profile) on reference planes.
- **Constraining**: Apply dimensional and geometric constraints so shapes behave predictably when parameters change.
- **Extrusion (Pad/Revolve)**:

- **Pad** the circle sketch to create the cylindrical body
- **Revolve** the profile to form a nose cone
- **Pad** the fin profiles and use **Polar Pattern** to repeat them

B. Boolean Operations

- Use **MultiFuse** or **Boolean Union** to combine multiple parts (body, nose cone, fins) into a single object.

C. Automated Assembly

- While FreeCAD doesn't have an official Assembly Workbench by default, you can use:
  - **A2plus** or **Assembly 4** (external workbenches) for proper mechanical assembly
  - Or simulate it with proper part positioning and fusing in the **Part Workbench**

D. Technical Drawing Generation

- **TechDraw Workbench** allows:
  - Projection of 3D views into 2D drawings
  - Adding precise dimensions and labels
  - Export to PDF/DXF for documentation or manufacturing

3. Technical Aspects of the Design

A. Units and Scale

- FreeCAD supports multiple unit systems. Ensure you're working in **mm** or **cm** depending on the model size.
- Use **Draft Scale** or **Drawing Templates** to fit large models (e.g., 2-meter projectile) into standard sheets (A3, A4).

B. Material Assignment and Rendering

- While FreeCAD doesn't focus on photo-realistic rendering, you can:
  - Apply colors and transparency for visualization
  - Export to Blender or POV-Ray for detailed rendering

C. File Outputs

- Save designs in FreeCAD's native .FCStd format.
- Export to:
    - **STEP, IGES** for industry-standard CAD exchange
    - **STL** for 3D printing
    - **DXF** for laser cutting or CNC machining
    - **PDF** for documentation

4. Projectile Model Specifics

Feature	**Value**
Length	1100 mm
Diameter	150 mm
Nose Cone	500 mm
Fins	4 fins, 200×300 mm, 45° sweep

- Nose cone: Revolved sketch forming aerodynamic shape
- Fins: Sketched once and replicated with **PolarPattern**
- All parts fused to form a single printable or machinable body

